

A Real-Time Traffic Digital Twin from Public CCTV: Vision-Based Detection, Map-Anchored Localization, and Self-Calibrating Speed Estimation

Minsoo Ahn

Department of Computer Science
Stony Brook University
minsoo.ahn@stonybrook.edu

Dahyun Kwon

Department of Computer Science
Stony Brook University
dahyun.kwon@stonybrook.edu

Abstract—We present a real-time traffic digital twin that converts live public CCTV feeds into an interactive, map-anchored view of road traffic. From a single clicked camera the system streams HLS video, detects vehicles with YOLOv8 and tracks them with a pluggable multi-object tracker (BoxMOT), and converts each vehicle’s pixel position to a geographic coordinate through a perspective homography. Because public CCTV cameras are uncalibrated and their intrinsics are unknown, accurate ground-plane mapping and speed estimation are the central challenges. We address them with three complementary mechanisms: (i) a two-stage *curved* transform that maps pixels along the true road centreline obtained from the national node-link road network rather than a flat plane; (ii) a five-stage speed estimator combining a physics-based outlier guard, least-squares regression over a sliding window, and per-track exponential smoothing; and (iii) an online self-calibration loop that compares the measured speed against the official ITS segment speed and learns a per-camera correction factor. The system additionally provides multi-camera background monitoring, camera-level congestion clustering, and a SQLite time-series history with charting. We describe the architecture and the key algorithms, and report latency, throughput, tracking-stability, and speed metrics produced by a measurement harness over live streams.

Index Terms—digital twin, intelligent transportation systems, object detection, multi-object tracking, homography, perspective transform, speed estimation, real-time visualization.

I. INTRODUCTION

Cities expose thousands of public traffic cameras, but the raw feeds are difficult to interpret at scale: an operator watching a video grid cannot read off vehicle counts, speeds, or where congestion is forming on a map. A *digital twin*—a live, geo-referenced model of the road state—turns those feeds into actionable information. The technical obstacle is that public CCTV cameras are *uncalibrated*: their height, pitch, focal length, and exact heading are unknown, so projecting an image detection onto the ground plane (and therefore measuring distance and speed) is ill-posed without per-camera calibration.

This paper describes a system that ingests Korean ITS CCTV streams and produces a real-time map-anchored twin. Our contributions, from a systems and algorithms standpoint, are:

- **Map-anchored localization.** We snap each camera onto the national node-link road network, extract the road centreline, and map pixels along the *curve* of the road

in two stages, rather than assuming a flat homographic plane. This keeps vehicle markers on the road even on bends.

- **Robust speed estimation without ground truth.** A five-stage filter (duplicate skip, physics jump guard, sliding-window least-squares regression, per-track exponential smoothing with spike rejection, and scale correction) produces stable speeds despite detection jitter and tracker ID churn.
- **Online self-calibration.** Because absolute scale is unknowable from an uncalibrated camera, we learn a per-camera `speed_scale` by comparing our measured average against the ITS official segment speed, with convergence and volatility guards.
- **A complete, resilient pipeline.** Automatic lane-based calibration, manual 4-point calibration, dual-path inference (browser canvas and server-side HLS) over a shared detector, multi-camera background monitoring, congestion clustering, and a historical analytics store.

The rest of the paper is organized as follows. Section II reviews related work. Section III presents the system architecture. Section IV details the algorithms. Section V reports the measurements. Section VI discusses limitations, and Section VII concludes.

II. RELATED WORK

Object detection and tracking. Modern traffic vision builds on real-time detectors; we use YOLOv8 [1] for its speed/accuracy balance and TensorRT FP16 execution [2]. For tracking we use the BoxMOT framework [3], which exposes ByteTrack [4], BoT-SORT [5], and OC-SORT [6]; appearance re-identification uses OSNet [7]. Trackers without appearance ReID suffer ID switches under occlusion, which we mitigate with a lightweight position-based ID stabilizer.

Camera-to-ground mapping. Mapping image points to a ground plane is a homography problem [8], typically solved from four correspondences with OpenCV [9]. For *uncalibrated* cameras, vanishing-point and lane-geometry cues are commonly used to recover pose. Our work combines a 4-point manual mode, a vanishing-point/lane auto-calibration, and—novel in

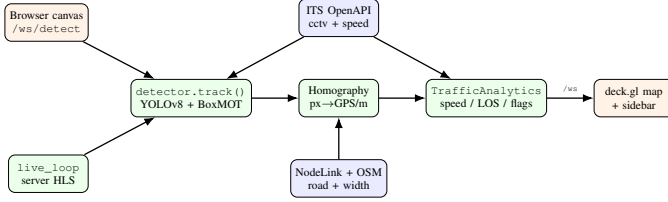


Fig. 1. System architecture. The browser path (`/ws/detect`) and the server path (`live_loop`) share one detector; a counter prevents concurrent tracker calls. External data (ITS CCTV/speed, NodeLink, OSM) feeds localization and self-calibration.

this context—a road-centreline-following transform driven by an external road-network dataset [10].

ITS dashboards and data. National ITS portals [11] publish CCTV streams and aggregated segment speeds, but typically as separate views. We *fuse* the segment speed back into the pipeline as a calibration signal. Congestion grouping uses density-based clustering [12] and convex hulls [13]; service levels follow the Highway Capacity Manual [14]. The visualization uses deck.gl [15] over MapLibre [16], served by FastAPI [17].

III. SYSTEM DESIGN AND ARCHITECTURE

The backend (FastAPI/Uvicorn) and a React/deck.gl frontend communicate over REST and WebSocket. Two inference paths can produce per-frame analytics and *share a single detector*; a connection counter coordinates them so that the multi-object tracker is never driven by two interleaved frame streams.

Browser path (`/ws/detect`). The frontend plays the HLS stream through a CORS proxy, captures canvas frames (JPEG, ≤ 640 px, ≈ 33 ms interval, at most two in flight), and sends them to the backend, which runs detection, tracking, localization, and analytics, then returns an annotated frame and broadcasts the analytics JSON to all clients.

Server path (`live_loop`). The backend can also read the HLS stream directly with OpenCV/FFmpeg and run the same pipeline. It is gated twice: it skips tracking while a browser detection client is active (to protect the shared tracker), and it skips all GPU work when no one is watching (the frontend reports tab visibility).

Background services. Periodic loops refresh expiring HLS tokens (30 min), poll the ITS segment speed for self-calibration (5 min), and snapshot all monitored cameras into a SQLite history while recomputing congestion clusters (30 s). The resulting map view is shown in Fig. 2.

IV. IMPLEMENTATION

A. Detection and Tracking

The detector loads the fastest available backend—TensorRT FP16 engine, then ONNX Runtime, then PyTorch—and exports `.pt` $\rightarrow$ `.engine` on first run when a GPU is present. A tracker *tier* is chosen by GPU memory: ByteTrack (CPU/low-VRAM), OC-SORT (occlusion-robust, no ReID), or BoT-SORT/DeepOC-SORT with OSNet appearance ReID on larger GPUs. For the non-ReID tiers we add two robustness layers: `_dedup_`

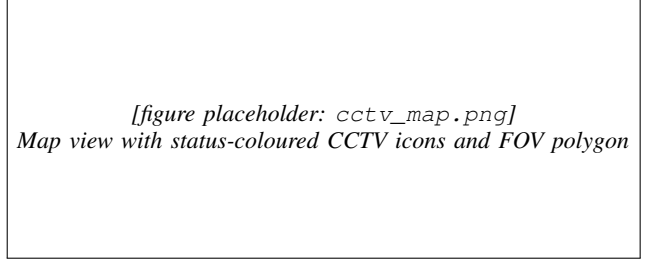


Fig. 2. Map view: public CCTV cameras as status-coloured icons on the vector map; the selected camera shows its field-of-view polygon and the located vehicles.

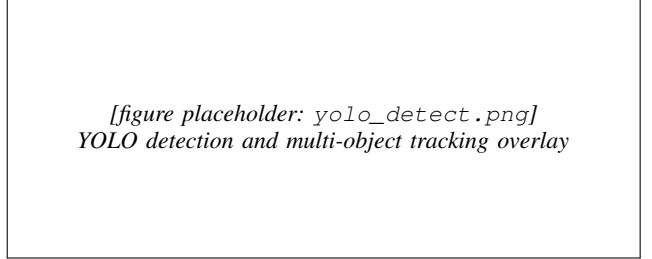


Fig. 3. Detection and tracking on a live CCTV frame: bounding boxes, vehicle class, and stabilized track IDs.

`tracks` merges duplicate boxes (IoU > 0.3 or centre distance < 40 px, keeping the older ID), and an `IDStabilizer` re-binds a vehicle's previous ID after a brief miss using nearest-centre matching (≤ 80 px) with a two-pass scheme and stale-mapping purge so display IDs stay stable and compact. Fig. 3 shows the detection and tracking overlay.

B. Map-Anchored Localization

On camera switch we query the national node-link network: we snap the camera position perpendicularly onto the nearest road polyline, extract ± 150 m of centreline, and—because each direction is stored as a separate one-way link—we locate the reverse carriageway and average the two snaps (when 2–40 m apart) to recover the true road centre. The chosen road's speed limit, lane count, and width (lanes $\times 2 \times$ lane-width) are attached to the camera; OSM Overpass provides a width refinement when available.

Given the centreline, the *two-stage curved transform* maps a pixel (u, v) to GPS along the road (Algorithm 1): stage one maps the pixel to local metres via the metric homography; stage two decomposes the displacement from the snap point into along-road and lateral components, advances that arc length along the centreline, interpolates the on-road position, and re-applies the lateral offset perpendicular to the *local* road bearing. This follows real road curvature, unlike a single flat homography.

When no manual calibration exists, an automatic procedure estimates a homography from one frame: Canny edges in the lower image, Hough line filtering, multi-row sampling of left/right road edges, least-squares edge fits, vanishing-point intersection, a pitch/height estimate from a pinhole model, and a trapezoid-to-GPS homography. Direction (which way the

Algorithm 1 Two-stage curved pixel→GPS transform

Require: pixel (u, v) ; metric homography H_m ; centreline $road_pts$; $snap_along$; direction sign s

- 1: $(x_m, y_m) \leftarrow H_m \cdot (u, v)$
- 2: $(dx, dy) \leftarrow (x_m, y_m) - snap_m \quad \triangleright$ ENU from snap
- 3: $d_{\parallel} \leftarrow dx \sin b + dy \cos b; \quad d_{\perp} \leftarrow dx \cos b - dy \sin b$
- 4: $arc \leftarrow snap_along + s \cdot \max(0, d_{\parallel})$
- 5: $(lat, lon) \leftarrow \text{INTERPCENTERLINE}(arc)$
- 6: $b_{\ell} \leftarrow \text{ROADBEARINGAT}(arc)$
- 7: **apply** $d_{\perp}$ perpendicular to b_{ℓ} **return** (lat, lon)

[figure placeholder: auto_calib.png]
Automatic lane-based calibration result

Fig. 4. Automatic calibration: detected lane edges and the estimated vanishing point yield the ground homography and the field-of-view footprint.

camera looks along the road) is resolved by matching the *image* road-curve sign against the *map* curve sign, falling back to a vanishing-point heading test. A manual 4-point mode remains available for maximum accuracy. Fig. 4 shows an automatic calibration result.

C. Speed Estimation

Speed is distance over time, and both terms are error-prone: homography distance is noisy at the far field, and HLS buffering makes naive frame timing wrong. We use a monotonic clock derived from stream presentation timestamps (falling back to wall-clock), and the five-stage estimator of Algorithm 2 with the constants in Table I.

D. Online Self-Calibration

The absolute distance scale of an uncalibrated camera is unknowable, so a residual scale error persists after geometric calibration. Every five minutes we fetch the ITS official segment speed near the camera and update a per-camera factor `speed_scale`:

$$target = scale \cdot \frac{v_{ITS}}{\bar{v}_{ours}}, \quad scale \leftarrow (1-\alpha) scale + \alpha target, \quad (1)$$

clamped to $[0.3, 5.0]$, using a 10-minute rolling average (≥ 50 samples), a coefficient-of-variation guard (≤ 0.4) to skip volatile traffic, and a learning rate that drops from 0.5 to 0.95 once converged (three consecutive $< 1\%$ updates). The converged factor is persisted per camera.

E. Aggregation, Congestion, and History

A horizontal line zone yields In/Out counts and per-vehicle direction. Dwell counting flags bottlenecks and parked vehicles (the latter remembered by pixel position so they survive ID changes), and a Level-of-Service grade is assigned from the

Algorithm 2 Per-track speed estimation (`analytics._speed`)

- 1: **(1)** skip duplicate metric positions (non-detection frames)
- 2: **(2)** **if** $step > (\frac{V_{max}}{3.6\sigma}) dt \cdot 1.5$ **then** drop sample, keep window
- 3: **if** $dt > 2s$ **then** clear window
- 4: **(3)** fit velocity by least squares over the $W=18$ -frame window;
- 5: **if** window displacement $< 0.5m$ **then** $v \leftarrow 0$
- 6: **(4)** EMA: $v_e \leftarrow (1-\alpha)v_e + \alpha(v\sigma)$, $\alpha=0.35$;
- 7: reject spike if $v_e > 5 \wedge v\sigma > 2.5v_e + 20$; decay when stopped
- 8: **(5)** output v_e ; **is_speeding** $\leftarrow v_e > limit$

TABLE I
KEY SPEED/ANALYTICS CONSTANTS (`CONFIG.PY`).

Parameter	Value
Speed window W	18 frames
EMA α	0.35
Jitter threshold	0.5 m
Min speed (else 0)	5 km/h
Max plausible V_{max}	180 km/h
GC grace	30 frames
Bottleneck dwell	150 frames (≈ 5 s)
Parked dwell	300 frames (≈ 10 s)
LOS thresholds A–E	$\leq 3, 6, 9, 12, 15$ (else F)

active vehicle count. Background monitoring polls additional cameras every 8 s with detection only; busy/congested cameras are clustered with a haversine density-based scan ($\varepsilon=500$ m) and outlined with a convex hull. A WAL-mode SQLite store records 30 s snapshots (14-day retention) and serves bucketed time series, peak-time detection, and CSV export. Fig. 5 shows the congestion overlay.

F. Visualization

The deck.gl map layers congestion polygons, vehicle trails, the FOV polygon, status-coloured camera icons, and vehicle markers. The FOV polygon uses one of three strategies in priority order: the manually clicked GPS ring; a curved corridor that follows the road centreline (mirroring the backend transform); or, when uncalibrated, a default trapezoid.

V. EVALUATION

Measurements are produced from the real pipeline in two ways. While the system runs (`make dev`), an in-process collector records per-frame stage latency, tracking, speed, and detection statistics and flushes `eval_*.csv` and `eval_summary.json` every 30 s (also on demand via `GET /eval/report`); an offline harness (`backend/evaluate.py`) reproduces the same metrics over a fixed clip for controlled benchmarks. Numbers below are produced on the target hardware.

Setup: model [YOLOv8x/s], backend [TensorRT/ONNX/PyTorch], tracker [tier], GPU [device], source [camera/clip], [N] frames.

[figure placeholder: *congestion_map.png*]
Congestion overlay: CCTV icons recoloured by status

Fig. 5. Camera-level congestion: monitored cameras are recoloured (normal/busy/congested) and adjacent congested cameras are grouped into a severity-shaded region.

TABLE II
PER-STAGE LATENCY (MS) AND THROUGHPUT. FILL FROM EVAL_
LATENCY.CSV.

Stage	Mean	Median	p95
YOLO (isolated)	—	—	—
Track (YOLO+BoxMOT)	—	—	—
Transform	—	—	—
Analytics	—	—	—
End-to-end	—	—	—
Throughput: — FPS			

Speed accuracy. Because no per-vehicle ground truth exists for public CCTV, we assess speed against the ITS official segment speed for the same camera and time window. Table IV compares the measured 10-minute average against ITS and reports the learned scale factor and convergence.

VI. DISCUSSION AND LIMITATIONS

The homography is accurate inside the calibration quadrilateral and extrapolates with growing error toward the far field; we mitigate this with the dual-matrix calibration, the regression window, lane-based auto-calibration, and the ITS `speed_scale`, but a residual error remains on poorly observed cameras. Automatic calibration assumes a focal length of $1.2 \times$ image height (unknown intrinsics), giving a $\pm 20\text{--}30\%$ scale uncertainty, and fails when lane markings are not visible (night, rain, dense traffic), falling back to an approximate grid. Road width is inferred from lane count and assumed bidirectional, so one-way roads are over-wide. The nearest node-link road is occasionally the wrong one (e.g., a camera near both a national route and an expressway), which displaces the FOV polygon and vehicle GPS; the CCTV-name road-hint reduces but does not eliminate this. Finally, congestion is clustered at the camera level because background cameras expose only counts, not per-vehicle speeds.

VII. CONCLUSION

We built a real-time traffic digital twin that localizes vehicles on a map from uncalibrated public CCTV and estimates their speed without ground truth. The key ideas are a road-centreline-following two-stage transform, a robust five-stage speed estimator, and an online self-calibration loop driven by official ITS segment speeds, wrapped in a resilient dual-path pipeline with congestion analytics and a historical store. The

TABLE III
TRACKING STABILITY AND SPEED DISTRIBUTION. FILL FROM EVAL_
TRACKING.CSV / EVAL_SPEED.CSV.

Metric	Value
Frames processed	—
Unique tracks	—
ID appearances (switch proxy)	—
Mean track lifetime (frames)	—
Median measured speed (km/h)	—
% moving observations	—

TABLE IV
MEASURED VS. ITS SEGMENT SPEED (PER CAMERA). FILL FROM THE LIVE
OUR_AVG_KPH/ITS_SPEED_KPH/SPEED_ERROR_PCT BROADCAST AND
SPEED_SCALE.JSON.

Camera	Ours (km/h)	ITS (km/h)	Err %	scale	conv.
—	—	—	—	—	—
—	—	—	—	—	—

approach makes raw municipal camera feeds directly usable for situational awareness, and its self-calibration removes the main barrier—per-camera manual setup—to scaling across many cameras.

VIII. CONTRIBUTION

This section states, per author, what was already available, what each of us specifically added, why it is technically different, and the evidence in the codebase. The focus is technical novelty rather than effort percentages. The two authors co-designed the architecture and integrated their parts; the split below reflects primary ownership.

A. Minsoo Ahn — Detection/Tracking, Analytics, and System Infrastructure

What was already there. YOLOv8 [1] and BoxMOT [3] provide detection and tracking; a basic distance/time formula gives speed. Out of the box these produce ID churn on non-ReID trackers and unstable, spiky speeds.

What I added.

- *Robust speed estimation* (`analytics.py:_speed`): the five-stage filter of Algorithm 2 (physics jump guard that drops only the outlier sample, sliding-window least-squares, per-track EMA with spike rejection and stop decay), plus a PTS-first monotonic time axis (`main.py:_speed_timestamp_ms`) that fixed speeds collapsing to zero under HLS buffering.
- *LOS, bottleneck, and parked-vehicle logic* (`analytics.py`): dwell counting, position-memory parked detection that survives ID changes, and HCM-style service grades.
- *Detection/tracking robustness*: tracker-tier selection, the `IDStabilizer` (two-pass remap with stale-mapping purge), `_dedup_tracks`, and the empty-detection grace that preserves tracker state (`detector.py`).
- *Road network + backend infrastructure*: node-link snapping and the bidirectional centre fix (`nodelink.py`), the

FastAPI server with the dual-path shared-detector coordination, the ITS self-calibration loop, congestion clustering (congestion.py), the SQLite history (history.py), and background monitoring (main.py). On the front-end, the analytics panels, history charts, application state, WebSocket layer, and map data wiring.

Why it is technically different. Naive speed from frame indices is wrong under streaming jitter; my multi-stage estimator plus PTS timing and the online ITS speed_scale produce stable speeds *without ground truth*. The ID stabilizer recovers identities that non-ReID trackers lose, and the shared-detector coordination lets two inference paths coexist safely.

Evidence. analytics.py (_speed, calibrate_from_its, dwell/LOS), detector.py (IDStabilizer, track), nodelink.py, main.py (live_loop, _speed_timestamp_ms, BackgroundMonitor), congestion.py, history.py; the detection overlay (Fig. 3), the congestion overlay (Fig. 5), and the measurement harness (evaluate.py).

B. Dahyun Kwon — Camera Geometry, Calibration, and Map Visualization

What was already there. Off-the-shelf parts provided a flat 4-point homography (OpenCV findHomography) and a generic map renderer. These assume a calibrated camera and a planar ground patch, which public CCTV does not provide.

What I added.

- *Two-stage curved pixel→GPS transform* (transform.py:_pixel_to_gps_curved, set_road_corridor): pixels are mapped along the real road centreline (from the node-link network) instead of a flat plane, with along/lateral decomposition and arc-length interpolation (Algorithm 1).
- *Single-frame automatic calibration* (transform.py:auto_calibrate_from_frame): Canny→Hough→multi-row edge fit→vanishing point→pinhole pitch/height→trapezoid homography, with image-vs-map curvature matching to resolve viewing direction.
- *Manual 4-point calibration UI and the map visualization*: the calibration overlay with node-snapping (CalibrationMode.jsx), the ROI polygon editor (RoiEditor.jsx), and the deck.gl FOV polygon strategies including the curved corridor (MapView.jsx:computeRoadCorridorPolygon).

Why it is technically different. Standard homography mapping breaks on curved roads and at the far field; my centreline-following transform and the lane-geometry auto-calibration let an *uncalibrated* camera be localized without manual setup, and the front-end polygon is computed to mirror the backend geometry so the displayed coverage matches the actual mapping.

Evidence. transform.py (_pixel_to_gps_curved, auto_calibrate_from_frame, update_from_calibration); MapView.jsx FOV strategies;

CalibrationMode.jsx, RoiEditor.jsx; the automatic calibration result (Fig. 4) and the map view (Fig. 2).

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